

## View results

Respondent

46

Anonymous

226:17

Time to complete

### A. General information about the respondent

1. \* Country

Armenia



2. \* Name of organization(s) or entity(s) responsible for the preparation of the report

Higher Education and Science Committee of the Ministry of Education, Science, Culture and Sports of the Republic of Armenia

3. \* Website of organization(s) or entity(s) responsible for the preparation of the report

www.hesc.am

4. \* Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

## 6. Please describe the role/mandate of your organization

The Higher Education and Science Committee is a state body functioning within the Ministry of Education, Science, Culture and Sports of the Republic of Armenia.

The scope of activities of the Committee includes the functions of the former Science Committee and the Supreme Certifying Committee of the RA MESCS, as well as the Department of Higher and Postgraduate Professional Education of the RA MESCS, ensuring the implementation of the state policy aimed at the cooperation and integration of the fields of science and higher education.

The objectives of the Committee are as follows:

- participation in strategic planning and policy development in the fields of higher education and science;
- implementation and evaluation of the programs in the field of higher education and science, as well as providing support to students;
- regulation of the fields of higher education and science and creation of an adequate legal framework for ensuring their functioning and improving their quality;
- ensuring the proper functioning and progressive development of higher education and science as a crucial factor in the development of the economy and national security, raising the country's competitiveness, educational and cultural development, as well as social progress;
- enabling citizens to exercise their human and civil rights to higher and postgraduate professional education;
- ensuring the compatibility of the degrees of higher and postgraduate professional education in the Republic of Armenia with international ones, as well as recognition of diplomas and diploma supplements issued in the Republic of Armenia in Europe and other foreign countries;
- promotion of international academic mobility for students;
- promoting and fostering academic freedom and autonomy of higher education institutions;
- ensuring the proper implementation of the procedures for awarding scientific degrees;
- assisting the process of qualification of scientific and academic personnel;
- preserving and developing the scientific and academic potential of the Republic of Armenia, creating an effective system for training of scientific and academic personnel;
- promotion of effective interaction and integration between science, education, and manufacturing.

7. \* Full name of officially designated contact person(s) who completed this survey

Astghik Melik-Karamyan

8. \* Position of officially designated contact person(s) who completed this survey

Head of the Unit for Public Relations and Information of the RA MESCS Higher Education and Science Committee

9. \* Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Harutyun Sargsyan, Deputy Chairman of the RA MESCS Higher Education and Science Committee

11. Brief description of the consultation process established for the preparation of the report

The report was based on the public information available on the websites of Armenian universities and research organizations, as well as the data provided in their annual activity reports.

12. \* Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

RA Ministry of Education, Science, Culture and Sports, Armenian universities and research organizations

13. \* Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey  
(Please use commas to separate multiple links, as: link A, link B)

<https://escs.am/am>, [ysu.am](https://ysu.am), [ysmu.am](https://ysmu.am), [brusov.am](https://brusov.am), [asue.am](https://asue.am)

14. \* Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

☒ Public

☐ Private

☐ Civil Society

☐ Other

## B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

### Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

## ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒ Yes

☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

With the coordination of the Ministry of Education, Science, Culture & Sport of Republic of Armenia, 4 Armenian universities (Armenian State University of Economics, Yerevan Brusov State University of Languages and Social Sciences (YSULS), Yerevan State Medical University, Yerevan State University (YSU) were involved in the "Strengthening Research Management and Open Science Capacities of HEIs in Moldova and Armenia (MINERVA)" Erasmus+ Program. Under the Program, representative of other research organizations and universities also took part in informational meetings and trainings.

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☐ Yes

☒ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

20. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Most of the research periodicals in Armenia are free of charge which contributes to accessibility of research results within the country. In addition, to foster the international accessibility of research result of Armenian researcher, a new grant program was launched by the Higher Education and Science Committee to reimburse the publication fee for Armenian researchers publishing their results in Open Access journals. At the same time, for all other 20 grant programs carried out by the Higher Education and Science Committee, the research groups are entitled to allocate budget means for publications in Open Access journals (indexed by Scopus and WoS)

21. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

22. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

There is a mandatory provision in the terms of the research grants launched by the Higher Education and Science Committee, urging the beneficiaries of the grant programs to ensure the creation and dissemination of popular science materials about the Project or their particular field of science (interviews, publications in popular science journals, creation, and distribution of video materials, organization of discussions for general audience, etc.). These dissemination efforts shall be considered when evaluating and accepting the Project's final report.



## Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

23. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☐ Yes

☒ No

24. 2.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The main strategy, principles and objectives for the development of the field of science are outlined in the Government 2021-2026 Program and the Action Plan derived thereof. There is no separate STI policy apart from that document.

25. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

☐ Yes

☐ Partly

☒ No

26. 2.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

The Roadmap for the integration of the field of R&D of the Republic of Armenia in the ERA (Order MESCS, 29 January 2020, N59) already includes the following open science activities – promotion of open access to scientific publications and data and supporting the development of knowledge transfer and open innovation. At the same time, according the ESCS Minister's Order N2121-A/2 dated 29 Dec 2021, an Expert Group for Open Science was established with the aim of studying the possibilities of OS full implementation in Armenia. Currently, the MESCS has elaborated the draft version of the "Goals and Guidelines on Open Science Policy of the RA" and the activities towards its adoption as a legal act will be carried out after the adoption of the new RA Law "On Higher Education and Science".

27. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☒ Yes
- ☐ Under development
- ☐ No

28. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

As mentioned above, in 2024, the Higher Education and Science Committee launched a new grant program with a total budget of approximately 1 million EUR to reimburse the publication fee for Armenian researchers publishing their results in Open Access journals thus fostering the international accessibility of their research findings.

29. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☐ Yes
- ☐ Under development
- ☒ No

30. 2.4 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

Currently, the MESCS has elaborated the draft version of the "Goals and Guidelines on Open Science Policy of the RA" and the activities towards its adoption as a legal act will be carried out after the adoption of the new RA Law "On Higher Education and Science".

31. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

32. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

Under the "Strengthening Research Management and Open Science Capacities of HEIs in Moldova and Armenia (MINERVA)" Erasmus+ Program, 4 Armenian universities (Armenian State University of Economics, Yerevan Brusov State University of Languages and Social Sciences (YSULS), Yerevan State Medical University, Yerevan State University (YSU) developed their own strategies for 2021-2026. Here are the relevant links:  
<http://www.old.ysu.am/science/en/1667990709>, <https://asue.am/en/https-/asue-am/am/-science>,  
[https://brusov.am/en/page\\_list/open\\_science/#sthash.VZJdsVd5.dpbs](https://brusov.am/en/page_list/open_science/#sthash.VZJdsVd5.dpbs),  
<https://ysmu.am/en/research/hanrahasaneli-gitutyun/>

33. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

☐

Yes

☐

Under development

☒

No

34. 2.6 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

There is no separate mechanism for open science, however OS fostering activities are supported in the frameworks of the three funding mechanisms of research and technical activities in the RA: baseline funding (facilities, labs, devices, salaries, maintenance), grant program funding and targeted (interdepartmental cooperation with other ministries). As it was mentioned before, there is a separate grant program for encouraging Armenian researchers' publications in Open Access journals. In addition, the current research periodicals in Armenia are Open Access journals.

35. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

36. 2.7 cont. **If yes**, please provide details and links as relevant.

In addition to the provisions mentioned in p. 35, the Higher Education and Science Committee has launched a portal <https://equipment.hesc.am/> where the details of all the research equipment of state research organizations is available.

37. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

38. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

After the adoption of Open Science White Paper in Armenia, it will be possible to take measures to foster public-private partnership in research field.

39. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☐ Yes
- ☐ Under development
- ☒ No

### Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

40. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

In 2024, the budget allocations for R&D comprised 0.39 % of the GDP of Armenia.

41. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

According to the information available at [armstat.am](https://armstat.am), in 2023, 80% of Armenian household members had access to internet connection (both landline and mobile). Find more at: [https://armstat.am/file/article/poverty\\_2024\\_a\\_4.pdf](https://armstat.am/file/article/poverty_2024_a_4.pdf)

42. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (*ref.: (iii) 18.c*)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

43. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

There is no official NREN network in Armenia, however there is Academic Scientific Research Computer Network of Armenia (asnet.am) that carries out some of the functions of such network.



44. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☒ Yes
- ☐ Under development
- ☐ No

45. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- ☐ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

46. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Under the "Strengthening Research Management and Open Science Capacities of HEIs in Moldova and Armenia (MINERVA)" Erasmus+ Program, 4 Armenian universities (Armenian State University of Economics, Yerevan Brusov State University of Languages and Social Sciences (YSULS), Yerevan State Medical University, Yerevan State University (YSU) developed their own scientific repositories. In addition, all types of academic theses (Bachelor's, Master's and PhD theses) are stored in open access by the National Library of Armenia.

47. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☐ Yes

☒ No

48. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

☒ Yes

☐ No

☐ Other

49. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☐ Yes

☒ No

## Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

### AND CAPACITY BUILDING FOR OPEN SCIENCE

50. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

51. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

The 4 beneficiary universities of "Strengthening Research Management and Open Science Capacities of HEIs in Moldova and Armenia (MINERVA)" Erasmus+ Program, continue sharing their knowledge and competence within their organization and other organization even after the implementation of the program.

52. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

53. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As OS implementation is relevantly new in Armenia, there are still no regular seminars for decision makers on that topic.

54. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

55. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As OS implementation is relevantly new in Armenia, there universities are still on their way to incorporate that topic in their respective curricula.

56. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☐ Partly
- ☒ No

57. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☐ Yes
- ☒ No

58. 4.5 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As the Open Science strategy is still on the way, more initiatives will be enabled after the adoption of the White Paper.

## Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

59. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒

Yes

☐

No

60. 5.1 cont. **If yes**, please provide more information and links as relevant.

The Armenia-based research periodicals are mostly open access and most of them has their own archives available. In addition, a certain part of Armenian researcher (depending on the field of science) initially publish their research on archive.org

61. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐

Yes

☒

No

62. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As the Open Science strategy is still on the way, more such initiatives will be enabled in the future.

63. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes

☒ No

64. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As the Open Science strategy is still on the way, more such initiatives will be enabled in the future.

65. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☒ Yes

☐ No

66. 5.4 cont. **If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

As an unintended collateral outcome of Open Science, Armenian scientists had problems with identifying predatory or fraud journals which caused them to publish their results in such journals. During the last few years, the Higher Education and Science Committee provides consultations to researchers on whether the journal is included in Beall's list (<https://beallslist.net/>) of predatory and fraud publications or journals, as well as discourages researchers from publishing their publications there by including a special provision in the Committee's Calls for proposals disqualifying the publications in the mentioned journals.

## Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

### OF THE SCIENTIFIC PROCESS

67. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☐ YES, at the national level
- ☐ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☒ No

68. 6.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As the Open Science strategy is still on the way, such initiatives will be enabled in the future.



## Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

69. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

In the majority of the Grant Programs carried out by the Higher Education and Science Committee, it is mandatory to involve a Foreign Collaborator, as well as there are bilateral grant programs for supporting joint research within international agreements.

70. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

71. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As the Open Science strategy is still on the way, another strategy on international cooperation is yet to be elaborated in the future after SWOT analysis in this field.

72. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

73. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is still need for a more profound research of potential collaborators and partners in this filed, after which it will be possible to have more tangible results.

## Part 8. GENERAL CONSIDERATIONS

74. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☒ Yes

☐ No

**75. 8.1 cont. If yes, please provide more information.**

Preliminary SWOT analysis of the research community and infrastructure has shown the following specific features. Weaknesses • Low level of open science awareness among researchers and lack of proper understanding of benefits of opening research data and protocols • Interdisciplinary approaches aren't practiced widely • Unwillingness to share data • Immature legislative basis for data protection • Insufficient use of diverse metrics in evaluation and review processes Threats • The lack of personnel capable to work with and maintain open science infrastructures • Underdeveloped state of semantic analysis tools for Armenian language and insufficient state of language technologies for Armenian language • Low level of trust in open science opportunities and inadequate appreciation of open science practices • Low prioritization of open science policy development in the R&D governance

**76. ANY OTHER COMMENTS:** If you wish, you may enter any additional comments or information here. You may also contact [openscience@unesco.org](mailto:openscience@unesco.org)

The preparation of this survey helped us to summarize the current state-of-the-art of Open Science in Armenia which will contribute to more efficient activities in this regard in the future.